

Near-linear Ryser equality families require superlinear edge counts

Research continuation, 7 October 2026. This proves a structural limit theorem for the same partite cover-number programme. The deduction uses a precisely cited published edge-colouring theorem. No priority claim, full Lean formalization or solution of Ryser's conjecture is asserted.

1. The structural conclusion

Throughout, H is a finite simple intersecting r -partite r -uniform hypergraph, $r \geq 2$, with m distinct edges and cover number $t = \tau(H)$. Each edge contains exactly one vertex of each specified part. Set

$$I(H) = \sum_{\{A,B\} \subset E(H)} (|A \cap B| - 1) \geq 0,$$

where the sum is over unordered pairs of distinct edges. Empty families have both cover number and excess zero. Linearity of an intersecting family is equivalent to $I(H) = 0$.

Theorem 1 (superlinear edge necessity). Let $r_j \rightarrow \infty$ and let H_j satisfy these hypotheses, with $\tau(H_j) \geq r_j - 1$ and $I(H_j) = o(r_j^2)$. Then

$$\frac{|E(H_j)|}{r_j} \longrightarrow +\infty. \quad (1)$$

In particular, no infinite family of linear intersecting partite hypergraphs with $\tau \geq r - 1$ can have a bounded edge/rank ratio.

Define $f_{\text{lin}}(r)$ as the minimum edge count in this linear class, with value $+\infty$ when the class is empty. Then

$$f_{\text{lin}}(r)/r \longrightarrow +\infty. \quad (2)$$

This gives no rate such as $r \log r$, and does not establish nonemptiness at all ranks. It does not determine the minimum in the larger nonlinear class or prove $\tau(H) \leq r - 1$.

The quantitative theorem behind this limit is the following. For every fixed integer $D \geq 4$, put

$$L = D + 1, \quad c_D = \frac{1}{2} + \sqrt{2D}, \quad y_D = \frac{(D-3)(D-2)}{2}, \quad A_D = \frac{L(y_D - 1)}{3}.$$

Theorem 2 (degree-parameter cover inequality). For every $D \geq 4$ and $\delta > 0$, a finite constant $C_{D,\delta}$ exists, independent of H, r , such that

$$m \geq L\tau(H) - (L - c_D + \delta)r - \frac{A_D I(H)}{r} - C_{D,\delta}. \quad (3)$$

Thus under $\tau(H) \geq r - 1$,

$$\frac{m}{r} \geq c_D - \delta - A_D \frac{I(H)}{r^2} - \frac{C_{D,\delta} + L}{r}. \quad (4)$$

The parameter D is fixed before taking $r \rightarrow \infty$. There is no claim of uniform control of $C_{D,\delta}$ as D grows. This order of quantifiers suffices for Theorem 1, because $c_D \rightarrow \infty$.

The new mechanism replicates a block of size $d \geq 4$ exactly $d - 3$ times and weights each copy by $(d - 2)/2$. This aligns the weighted matching gain with the star degree and an exact pair-count identity. The preceding notes use degrees at most four and weighted three/four blocks; the arbitrary fixed-degree replication here gives the superlinear conclusion. Degree peeling, pair-excess deletion and matching/star tools have predecessors in Sivashankar [S, Section 4].

2. The published input and parallel copies

We use Kahn's small-codegree edge-colouring theorem, in the form stated by Kang–Kelly–Kühn–Methuku–Osthus [KKKMO, Theorem 3.1]. For each fixed rank D and $\eta > 0$, there is $\gamma > 0$ such that a hypergraph of rank at most D , maximum degree at most a , and maximum pair codegree at most γa has list chromatic index at most $(1 + \eta)a$. Ordinary colouring is a consequence.

Their convention explicitly allows parallel edge copies: edges have identities, and distinct edges may have equal vertex sets. Two such copies cannot receive the same colour. Degrees and codegrees count copies. This convention is essential below and was checked in the primary manuscript, Section 1.2, as well as Theorem 3.1.

Choose an integer $a_0 = a_0(D, \eta) \geq 1$ so large that $D - 3 \leq \gamma a$ for all $a \geq a_0$. Thus Input K applies to rank at most D , codegree at most $D - 3$, and maximum degree $a \geq a_0$. We do not re-prove or formalize that published theorem. Every application below satisfies these hypotheses. No OpenAI/math theorem or degree-three residual lemma is needed for this note.

3. Peeling and an exact residual identity

Select a vertex of current degree at least $L = D + 1$, record it and delete its current incident edges. Continue until the residual J has maximum degree at most D . If k vertices are recorded and $q = |E(J)|$, then

$$m \geq q + Lk, \quad t \leq k + \tau(J), \quad m \geq q + L(t - \tau(J)). \quad (5)$$

It remains intersecting and r -uniform unless empty, and retains the original parts. Fixing a nonempty residual edge gives $q - 1 \leq r(D - 1)$; an empty residual also satisfies this. Therefore

$$q \leq (D - 1)r + 1 \leq Dr. \quad (6)$$

Let $T = I(J) \leq I(H)$; deleting edges only omits nonnegative summands. Write x_d for residual degree counts, $1 \leq d \leq D$, and define

$$W = \sum_{d=3}^D \frac{d-2}{2} x_d, \quad Y = \sum_{d=4}^D \frac{(d-3)(d-2)}{2} x_d, \quad K = \frac{q(q-r)}{2}.$$

Incidence and intersection counting give $\sum dx_d = qr$ and $\sum \binom{d}{2} x_d = \binom{q}{2} + T$. Subtracting half the incidence count gives the exact identity

$$3W + Y = K - q/2 + T + x_1/2. \quad (7)$$

The degree-one term has been retained with its correct sign; the degree-two term is zero.

Set $s = q/2 + 1 - \tau(J)$. Pairing the residual edges shows $s \geq 0$. In any fixed part, select every vertex of degree at least three. Their incident edge sets are disjoint. A degree- d selection uses one vertex in

place of the $d/2$ pairing cost, saving $(d-2)/2$. Pair the remaining edges, at a rounding cost at most $1/2$. Summing these savings over all parts gives W , so one part saves at least W/r . Hence

$$W \leq rs. \quad (8)$$

4. Linearize first, then replicate

View edges of J as q points. Each original residual vertex of degree $d \geq 4$ gives a d -point incidence block. Initially allow repeated blocks. Their pair codegrees λ_{AB} satisfy $\lambda_{AB} \leq |A \cap B|$, so

$$X = \sum_{\{A,B\}} (\lambda_{AB} - 1)_+ \leq T.$$

Whenever a pair has codegree at least two, delete one block containing it. The integer pair excess decreases by at least one and cannot increase elsewhere. After at most T such deletions the retained original-block graph is simple and linear, with sizes in $[4, D]$. Only auxiliary blocks are deleted here; original residual edges and the counts x_d stay fixed.

Give a size- d original block weight $y_d = (d-3)(d-2)/2$. Each deletion loses at most y_D , so the retained weight Y' satisfies

$$Y' \geq Y - y_D T. \quad (9)$$

This is valid even when its right-hand side is negative.

Now replicate each retained block of size d exactly $d-3$ times, giving each copy weight $w_d = (d-2)/2$. The total copy weight is precisely Y' . The replicated graph has rank at most D and maximum codegree at most $D-3$: every pair lies in at most one retained original block, with at most $D-3$ copies. Let a be its maximum degree, zero for an empty graph.

If $a \geq a_0$, Input K colours its copies with at most $(1+\eta)a$ colours. A colour class has weight at least $Y'/((1+\eta)a)$. It is a matching and contains no two copies of one original block. Select each class member's defining original vertex and pair all uncovered residual edges. The matching's cover saving is its sum of $(d-2)/2$, giving

$$s \geq Y'/((1+\eta)a). \quad (10)$$

For a second cover, choose a point of replicated degree a . Suppose it lies in n_d distinct retained size- d blocks. The other points of those original blocks are disjoint by linearity. Selecting their defining vertices covers exactly $1 + \sum (d-1)n_d$ edges at cost $\sum n_d$. Pairing the remainder gives

$$\tau(J) \leq q/2 - \frac{1}{2} \sum (d-3)n_d = q/2 - a/2, \quad s \geq a/2. \quad (11)$$

The star uses each original block once, not its parallel copies. Multiplying (10)–(11) yields $Y' \leq 2(1+\eta)s^2$.

If $a < a_0$, each copy has at least four points, so there are at most $qa_0/4$ copies. Each weight is at most $(D-2)/2$, giving $Y' \leq q(D-2)a_0/8$. Both cases and (9) imply

$$Y \leq y_D T + 2(1+\eta)s^2 + q(D-2)a_0/8. \quad (12)$$

In particular the colouring theorem is never used below its threshold.

Put $B = 1/2 + (D-2)a_0/8$. Combining (7), (8), (12) and $x_1 \geq 0$ gives

$$2(1+\eta)s^2 + 3rs \geq K - (y_D - 1)T - qB. \quad (13)$$

Since $D \geq 4$, $y_D - 1 \geq 0$.

5. The general scalar square and its errors

Define

$$z = \sqrt{2D}, \quad \alpha = \frac{D-1}{2(D+1)}, \quad d = 1 - \frac{1/2 + z}{D+1}, \quad \ell = \alpha q - dr, \quad \beta = \frac{1}{2} + \frac{D-1}{z}.$$

Here $0 < d < 1$ and $0 < \alpha < 1/2$. Indeed $(D + 1/2)^2 - 2D = (D - 1/2)^2 > 0$. The exact square identity is

$$K - (2\ell^2 + 3r\ell) = \frac{2D}{(D+1)^2}(q - \beta r)^2 \geq 0. \quad (14)$$

It can be verified by coefficient comparison using $z^2 = 2D$. The accompanying Lean file verifies a denominator-cleared identity for arbitrary real parameters and its implication.

Let $b = (1 + \eta)s$, $Z = (y_D - 1)T + qB$, and $E = Z/(3r)$. Since $r, s, \eta \geq 0$, (13) gives $2b^2 + 3rb \geq K - Z$. Equations (14) and this inequality imply

$$b + E \geq \ell. \quad (15)$$

Otherwise $\ell - b > E$ and $\ell > b \geq 0$, so

$$(2\ell^2 + 3r\ell) - (2b^2 + 3rb) = (\ell - b)(2(\ell + b) + 3r) > 3rE = Z,$$

a contradiction. A negative ℓ is covered automatically.

By (6), $\ell \leq Dr$ and $E \leq (y_D - 1)T/(3r) + DB/3$. If $s \leq Dr$, use $(1 + \eta)s \leq s + \eta Dr$ in (15); if $s > Dr$, the same conclusion is immediate. Thus $s \geq \ell - \eta Dr - E$. Rewrite (5) using the definition of s , and substitute this bound:

$$\begin{aligned} m &\geq Lt - (L/2 - 1)q + Ls - L \\ &\geq Lt - (L - c_D + LD\eta)r - \frac{L(y_D - 1)T}{3r} - \frac{LDB}{3} - L. \end{aligned} \quad (16)$$

The q terms cancel since $L\alpha = L/2 - 1$, and $Ld = L - c_D$. Set $\eta = \delta/(LD)$, replace T by $I(H)$, and take

$$C_{D,\delta} = L + LDB/3, \quad B = 1/2 + (D - 2)a_0(D, \delta/(LD))/8. \quad (17)$$

This proves Theorem 2, including empty families and residuals. The constants may be ineffective through the colouring input.

6. Consequences and quantifiers

For Theorem 1, fix any integer $D \geq 4$ and any $\delta > 0$. Under its hypotheses, (4) gives $\liminf m_j/r_j \geq c_D - \delta$. Letting $\delta \downarrow 0$ gives $\liminf m_j/r_j \geq c_D$. This holds for every fixed D ; their c_D are unbounded. Therefore $m_j/r_j \rightarrow +\infty$. No threshold is required to be uniform in D . For (2), any contrary subsequence with bounded $f_{\text{lin}}(r)/r$ would supply actual minimizing linear examples and contradict Theorem 1.

There is a quantitative complementary obstruction. If $\tau(H_j) \geq r_j - 1$, $r_j \rightarrow \infty$ and $m_j/r_j \rightarrow C < \infty$, then for each $D \geq 5$, (4) gives

$$\liminf I(H_j)/r_j^2 \geq \frac{(c_D - C)_+}{A_D}.$$

Taking the supremum over fixed integers $D \geq 5$ yields a strictly positive lower bound, because some $c_D > C$. No optimality is claimed. Thus every bounded-edge/rank family in this cover class must have quadratic intersection excess.

This conclusion distinguishes the linear class from the conjectured $f(r) = O(r)$ behaviour of the full partite class in [ABW, Conjecture 2.11]. It does not refute that full-class conjecture. The random truncated-projective-plane construction in [ABW, Theorem 1.3] gives a linear $O(r \log r)$ upper bound on suitable prime-power subsequences, consistent with our superlinear lower bound. Whether the correct linear growth is $r \log r$, or substantially smaller, is not settled here. The dated comparison report is a limited search, not a claim of priority.

There is also an explicit cover gap for a bounded edge budget.

Corollary 3 (bounded-budget cover gap). Fix a real $C \geq 1$. There is a constant K_C such that every linear hypergraph in our class with $m \leq Cr$ satisfies

$$\tau(H) \leq \left(1 - \frac{1}{2(C+1)}\right) r + K_C. \quad (18)$$

More generally the same bound holds with the extra term $(y_D - 1)I(H)/(3r)$, where $D = \lceil 2(C+1)^2 \rceil$.

Proof. Set this D , $L = D + 1$, and use Theorem 2 with $\delta = 1/2$. Since $2(C+1)^2 \leq D \leq 2(C+1)^2 + 1$, we have $c_D \geq 2C + 5/2$ and $L \leq 2(C+1)^2 + 2$. Consequently

$$\frac{c_D - C - 1/2}{L} \geq \frac{C + 2}{L} \geq \frac{1}{2(C+1)}.$$

The last inequality follows from $L/(2(C+1)) \leq C + 1 + 1/(C+1) \leq C + 3/2 < C + 2$. Rearrange (3) with $m \leq Cr$, and take $K_C = C_{D,1/2}/L$. The excess coefficient is $A_D/L = (y_D - 1)/3$, proving both statements. $\square$

References and verification

- [K] J. Kahn, *Asymptotically good list-colorings*, Journal of Combinatorial Theory, Series A 73 (1996), 1–59.
- [KKKMO] D. Y. Kang, T. Kelly, D. Kühn, A. Methuku and D. Osthus, *Solution to a problem of Erdős on the chromatic index of hypergraphs with bounded codegree*, Proceedings of the London Mathematical Society 129 (2024), no. 6, e70011, arXiv:2110.06181. Section 1.2 fixes the parallel-copy convention; Theorem 3.1 supplies the Kahn theorem. The primary statement was inspected.
- [S] V. Sivashankar, *An Improved Lower Bound for the Erdős–Lovász Cover Number Problem*, arXiv:2606.24878v2, Section 4, for the preceding peeling, linearization and matching/star framework. Its degree-three lemma is not needed by this theorem.
- [ABW] R. Aharoni, J. Barát and I. M. Wanless, *Multipartite hypergraphs achieving equality in Ryser’s conjecture*, Graphs and Combinatorics 32 (2016), 1–15, arXiv:1409.4833v2, DOI:10.1007/s00373-015-1575-9.

The standard-library checker verifies denominator-cleared polynomial identities, exact square-root constants and finite replication, matching and star diagnostics. These diagnostics are not the general proof. Partial Lean exports check the scalar implications, not the finite hypergraph reductions or the published colouring theorem. The audit states the exact scope, and no external mathematician has reviewed it.